Report 4. π0.5 zero-shot — from 98.4 % in MuJoCo to 61 % in Isaac Sim, with nearly half of the successes coming from regrasping

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0. What this report is about

To see whether a model outside the OpenVLA family also drops in the same way when moved from MuJoCo to Isaac Sim, we measured Physical Intelligence's π0.5 with the same evaluation code and the same criteria. π0.5 looks at the screen once and issues 10 actions. In these tests, as in openpi's LIBERO example, it executes the first 5 and then looks again.

We measured the public model trained on MuJoCo data without additional training (zero-shot), in 6 cells with 500 episodes each. The evaluation setup, test cells and criteria are described in Sections 1–4 of the README.

Conclusions at a glance

Question	Answer	Evidence
What happens when moved to Isaac Sim?	From 98.4 % in MuJoCo to 52.0–61.0 % in Isaac Sim. It is the highest of the three zero-shot models and retains 53–62 % of its MuJoCo performance	Section 2
How were those successes achieved?	Under Isaac M (MuJoCo-matched speed), 40–47 % of successes came **after losing the bowl and grasping it again** (3 % in MuJoCo). Counting only one-pass successes, Isaac M gives 30.0–34.6 %	Section 3
What if each action is executed for longer (B)?	The conclusion depends on the criterion. By the original criterion it drops slightly (−9.0 %p, z = −2.87 · −4.0 %p, z = −1.28). Counting only one-pass successes, it does not drop (+6.4 %p, z = +2.15 · +2.0 %p, z = +0.66). π0.5, executing 5 actions per chunk, is far less sensitive to execution-time differences than OFT with 8 (−33 %p)	Section 4
What if object placement changes?	Overall success does not change (−0.4 · +4.6 %p, z = −0.13 · +1.46)	Section 5
How does it fail?	Under M, failures are spread over all stages, and 17–27 % fail after bringing the bowl to the plate. Under B, more than half of the failures close the gripper outside the bowl	Section 6
Per task?	Tasks that work under M and under B are clearly different. Task 2 (center of the table) succeeds in 42–49 under B (21–25 under M), while task 5 (on the ramekin) succeeds in 34–35 under M and 0–1 under B. Task 7 (on the stove) is 0 under M	Section 7

1. Model and connection

Item	Details
Model	openpi `pi05_libero` (trained on LIBERO human demonstrations, no additional training)
Inputs	Front camera, wrist camera, 8 robot-state values (hand position 3 · orientation 3 · fingers 2), task sentence
Outputs	10 actions per observation; the first 5 are executed before the next observation (as in openpi's LIBERO example)
Front image	The central area 0.389 is cropped to match the MuJoCo training field of view. π0.5 was trained on uncropped MuJoCo images and does not crop inside the model, so the resulting view is the same as OpenVLA's and OFT's
Wrist image	Not cropped
Gripper value	π0.5's output (−1 open · +1 close) is converted to the evaluation code's format (0 close · 1 open)

Because π0.5 executes 5 actions in a row, it moves 5 times longer than OpenVLA before looking at the screen again.

	MuJoCo default	MuJoCo-B	Isaac M	Isaac B
Execution time per action	0.05 s	0.10 s	0.05 s	0.17 s
Until the next observation	0.25 s	0.50 s	0.25 s	0.85 s

The policy server (`code/pi05/pi05_server.py`) uses the same exchange format as the OFT server, so the Isaac Sim evaluation code uses its OFT-style connection unchanged. At start-up the server runs two dummy predictions. Otherwise the evaluation code would time out while the first prediction is being prepared. In MuJoCo, the gripper value is used as a continuous value, as in the openpi example.

2. Results

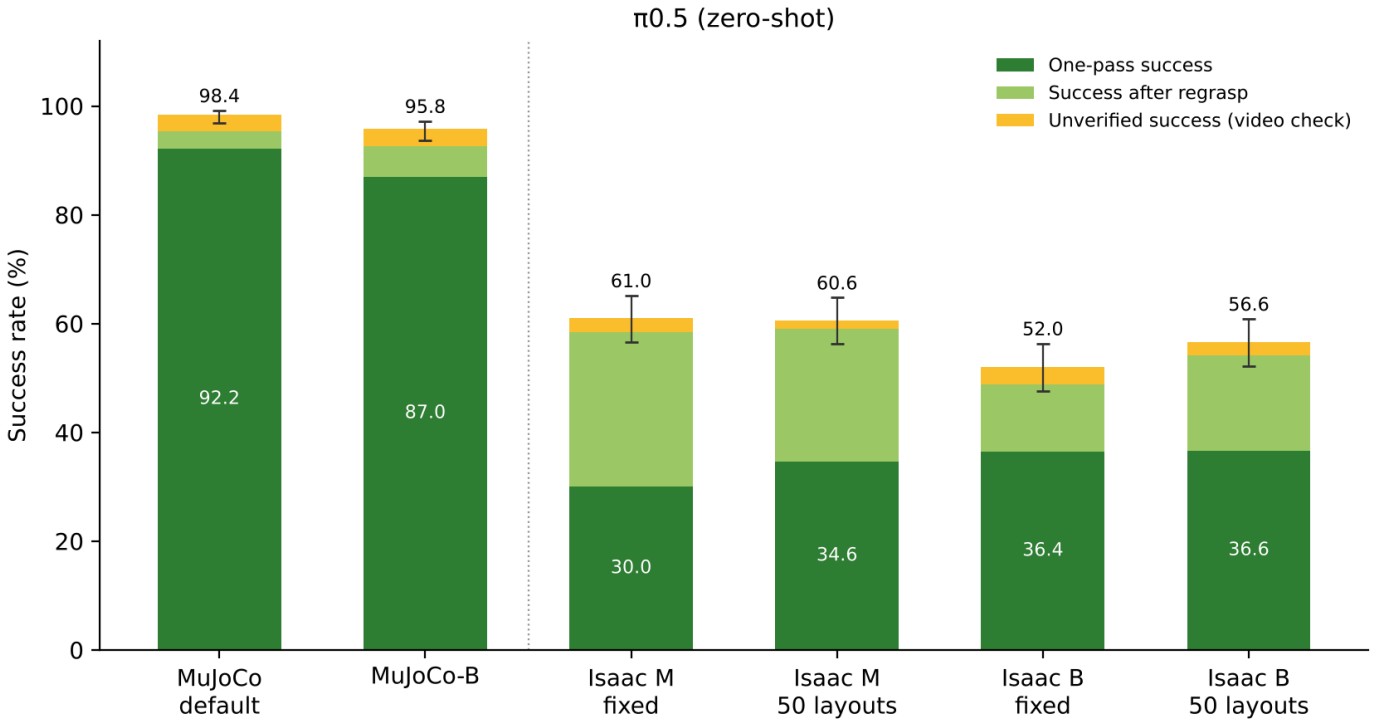


Figure 1. Success of π0.5. Dark green: one-pass success; light green: success after regrasp; yellow: unverified success. Lines above the bars: 95 % confidence range of success by the original criterion.

Cell	Success (original)	95 % confidence range	One-pass success	Retained vs MuJoCo default (original / one-pass)
MuJoCo default	492 (98.4 %)	96.9–99.2	461 (92.2 %)	—
MuJoCo-B	479 (95.8 %)	93.7–97.2	435 (87.0 %)	97 % / 94 %
Isaac M · fixed layout	305 (61.0 %)	56.7–65.2	150 (30.0 %)	62 % / 33 %
Isaac M · 50 layouts	303 (60.6 %)	56.3–64.8	173 (34.6 %)	62 % / 38 %
Isaac B · fixed layout	260 (52.0 %)	47.6–56.3	182 (36.4 %)	53 % / 39 %
Isaac B · 50 layouts	283 (56.6 %)	52.2–60.9	183 (36.6 %)	58 % / 40 %

Moving from MuJoCo default to Isaac M · 50 layouts costs 37.8 %p ($z = -14.80$). This is the smallest drop of the three zero-shot models, but even a model at 98 % in MuJoCo is around 60 % in Isaac Sim.

3. Successes obtained by grasping again

What we saw. In the test videos, π0.5 often lost the bowl or failed to grip it properly, then tried to grasp it again. The original criterion only checks whether the bowl ends up on the plate, so such episodes count as success just like clean one-pass episodes.

How we checked. For every episode we judged, from the finger-width and bowl/hand position logs, whether the bowl was carried without being lost (README Section 4).

Cell	Success (original)	One-pass	After regrasp	Unverified	Failure after a lost grasp	Other failure	Mean gripper closings per episode
MuJoCo default	492	461	16	15	2	6	1.06
MuJoCo-B	479	435	28	16	10	11	1.24
Isaac M · fixed layout	305	150	142	13	178	17	2.87
Isaac M · 50 layouts	303	173	122	8	175	22	2.80
Isaac B · fixed layout	260	182	62	16	207	33	2.88
Isaac B · 50 layouts	283	183	88	12	179	38	2.77

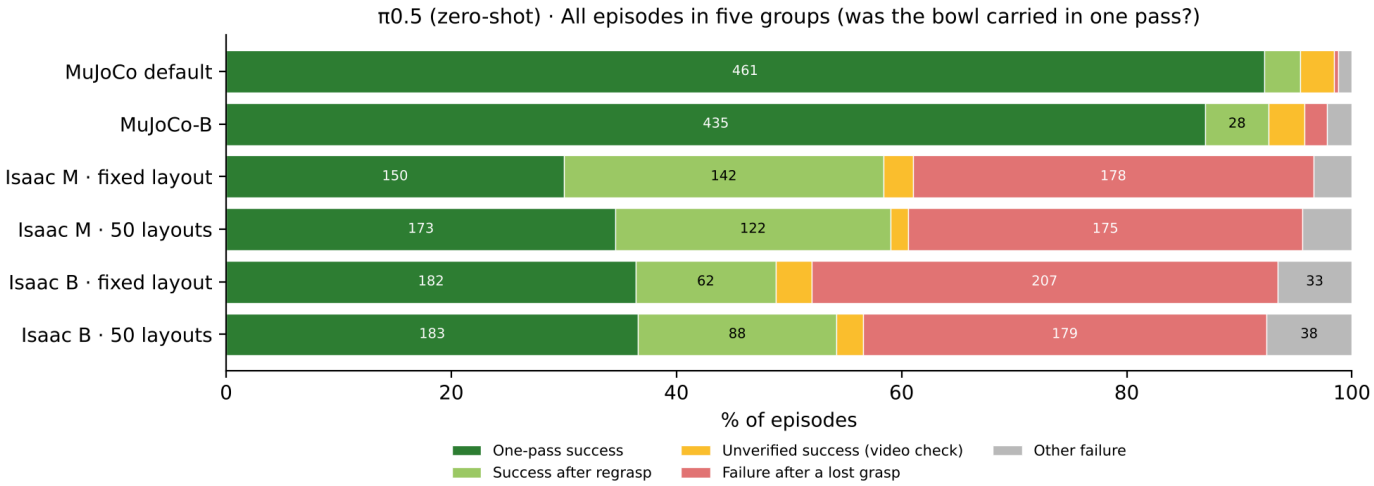


Figure 2. All 500 episodes in five groups (numbers are episode counts).

Results.

- **Under Isaac M, nearly half of the successes come from grasping again.** That is 142 of 305 (47 %) with the fixed layout and 122 of 303 (40 %) with 50 layouts. In MuJoCo default it is only 16 of 492 (3 %), so this behavior is new in Isaac Sim.
- **Most failed episodes also tried again.** In Isaac Sim, 82–91 % of failures are "failure after a lost grasp", and the gripper closes 2.8–2.9 times per episode on average (1.06 in MuJoCo).
- **Each cell has 8–16 unverified successes.** Under M, most "not held at release" episodes released the bowl 3–7 cm above the plate. That is just over the 3 cm threshold, closer to placing from slightly high than to dropping. Under B, two thirds of them released the bowl more than 10 cm above the plate, which is real dropping. "Never lifted with the hand" occurs in 9 episodes under M. Six of them are in task 8, where the bowl starts next to the plate. The other 3 are in task 9 (on the wooden cabinet), where the bowl starts higher than the plate, so dragging it down while holding it can trip this check even if it was carried properly (a limit of the check, to be confirmed on video).

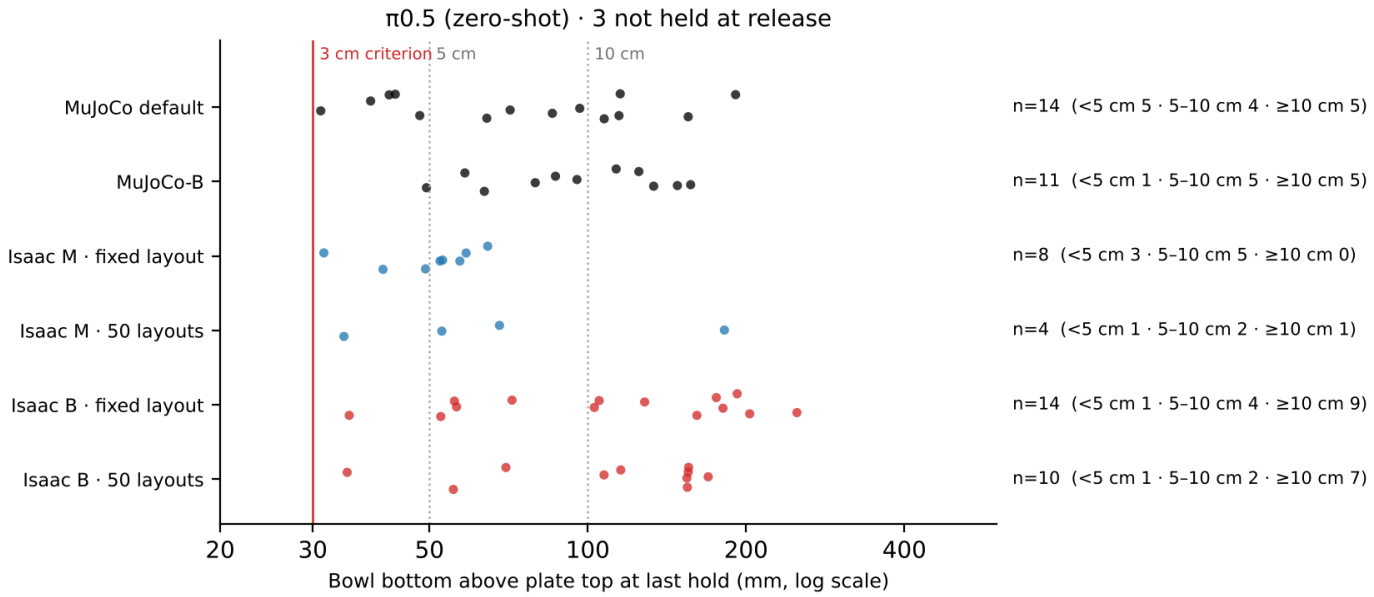


Figure 3. Release height of unverified successes marked "not held at release". The red line is the 3 cm threshold.

What we can say now. In Isaac Sim, π0.5 does not give up after a failed attempt; it tries again and raises its success rate that way. Comparing π0.5 with other models by the original criterion alone therefore overstates its ability to succeed in one go. Both criteria should be reported together.

4. Changing the execution setting (M → B)

Comparison	Original criterion	z	One-pass	z	One-pass: possible range once unverified are resolved (z)
Isaac M → B, fixed layout	61.0 → 52.0 % (−9.0 %p)	−2.87	30.0 → 36.4 % (+6.4 %p)	+2.15	+3.8 to +9.6 %p (+1.26 to +3.19)
Isaac M → B, 50 layouts	60.6 → 56.6 % (−4.0 %p)	−1.28	34.6 → 36.6 % (+2.0 %p)	+0.66	+0.4 to +4.4 %p (+0.13 to +1.44)
MuJoCo default → MuJoCo-B	98.4 → 95.8 % (−2.6 %p)	−2.45	92.2 → 87.0 % (−5.2 %p)	−2.69	—

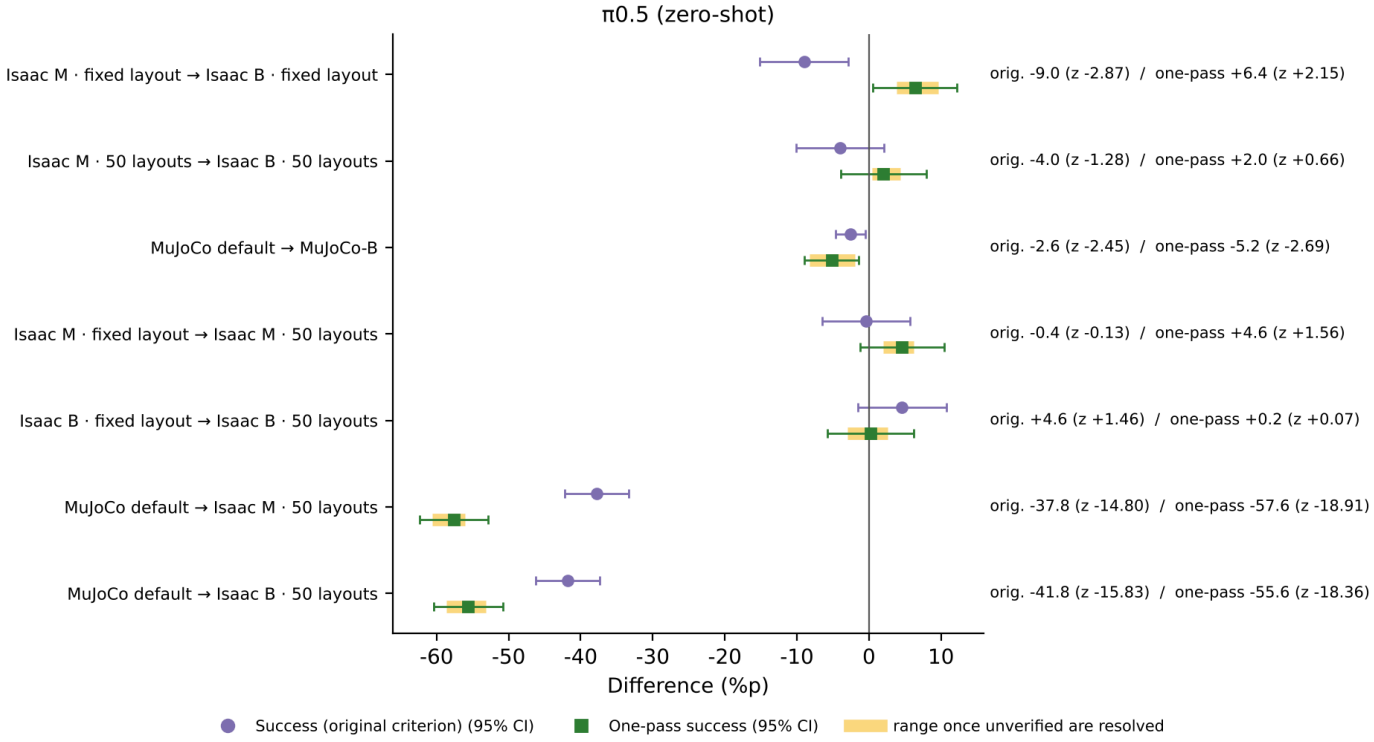


Figure 4. Differences when one factor is changed for π0.5. Circles: original criterion; squares: one-pass success; yellow band: possible range once unverified successes are resolved.

- **The order of M and B depends on the criterion.** By the original criterion M is higher, but counting only one-pass successes B is not lower than M. M has more successes because it regrasps more (M 122–142, B 62–88). With the fixed layout, the conclusion "B is higher" (z = +2.15) may become non-significant depending on how unverified successes resolve (z = +1.26). In no case, however, does M become higher than B.
- **Why M regrasps more (hypothesis).** Under M, failures where the bowl is gripped but not lifted are more common than under B (60–72 vs 26–29). This points the same way as the measurement that lifting under M moves 0.63× of MuJoCo (Section 4 of Report 2). We think episodes that lose the bowl while lifting and grasp it again become more common under M. We will check this with a setting that also matches the vertical motion (M2).
- **Effect of chunk length.** When the same change was made in MuJoCo, π0.5 dropped only −2.6 %p, while OFT, with 8-action chunks, dropped −33.4 %p (Report 3). The shorter the chunk, the less sensitive the model is to execution-time differences.

5. Changing the object placement (fixed layout → 50 layouts)

Setting	Success (original)	z	One-pass	z
M	61.0 → 60.6 % (−0.4 %p)	−0.13	30.0 → 34.6 % (+4.6 %p)	+1.56
B	52.0 → 56.6 % (+4.6 %p)	+1.46	36.4 → 36.6 % (+0.2 %p)	+0.07

Overall success does not depend on object placement. Per-task changes are also smaller than OFT's. Under M, apart from task 1 (32 → 23) and task 4 (7 → 18), most tasks change by 5 or fewer.

6. Failure kinds

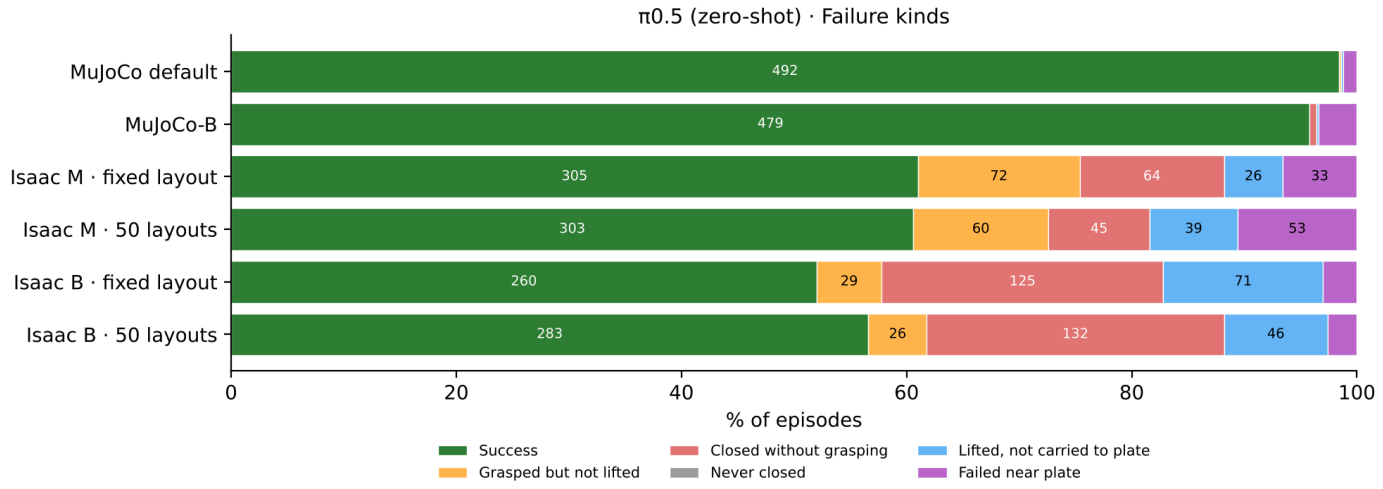


Figure 5. Failure kinds (share of episodes; numbers are episode counts).

Cell	Grasped but not lifted	Closed without grasping	Lifted, not carried to plate	Failed near plate
Isaac M · fixed layout	72	64	26	33
Isaac M · 50 layouts	60	45	39	53
Isaac B · fixed layout	29	125	71	15
Isaac B · 50 layouts	26	132	46	13

"Never closed" is 0 in every cell.

- **Under M, failures are spread evenly across the stages.** Episodes that bring the bowl to the plate and still fail ("failed near plate") are 17–27 % of failures, unlike OFT (0).

- **Under B, failures concentrate in the grasping stage.** Closing outside the bowl accounts for 52–61 %. We think the hand overshoots the grasp position during the 5-action chunk.

7. Per task

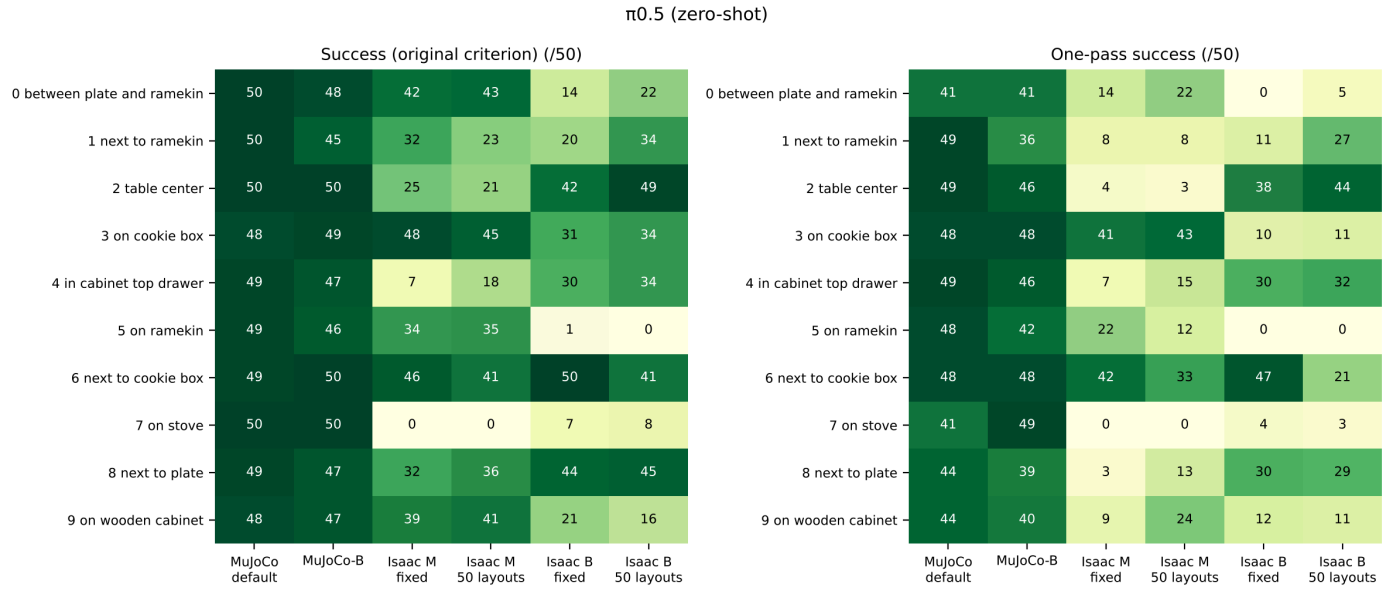


Figure 6. Successes per task × cell (out of 50). Left: original criterion; right: one-pass.

Task (where the bowl starts)	M · 50 layouts: original (one-pass)	B · 50 layouts: original (one-pass)	B − M (original)
0 between plate and ramekin	43 (22)	22 (5)	−21
1 next to ramekin	23 (8)	34 (27)	+11
2 center of the table	21 (3)	49 (44)	+28
3 on the cookie box	45 (43)	34 (11)	−11
4 in the cabinet top drawer	18 (15)	34 (32)	+16
5 on the ramekin	35 (12)	0 (0)	−35
6 next to the cookie box	41 (33)	41 (21)	0
7 on the stove	0 (0)	8 (3)	+8
8 next to the plate	36 (13)	45 (29)	+9
9 on the wooden cabinet	41 (24)	16 (11)	−25
Total	303 (173)	283 (183)	−20

- **Overall, M → B differs by only 20 episodes, but individual tasks split sharply.** Under B, tasks 2 (+28), 4 (+16) and 1 (+11) rise, while tasks 5 (−35), 9 (−25) and 0 (−21) fall.
- **Under M, task 2 succeeds by regrasping.** Under M only 3 of its 21 successes are one-pass; under B, 44 of 49 are.
- **Task 7 (on the stove) is 0 under M,** against 50 in MuJoCo. All three models trained only in MuJoCo hardly solve this task in Isaac Sim.

8. Detailed metrics

Cell	Grasp	Median placement error	Within 30 mm of plate center	Within 50 mm
MuJoCo default	100.0 %	12.1 mm	98.4 %	99.2 %
MuJoCo-B	99.2 %	19.7 mm	96.0 %	98.4 %
Isaac M · fixed layout	86.8 %	41.9 mm	37.4 %	56.4 %

Cell	Grasp	Median placement error	Within 30 mm of plate center	Within 50 mm
Isaac M · 50 layouts	90.0 %	42.9 mm	34.4 %	57.0 %
Isaac B · fixed layout	74.2 %	61.0 mm	25.6 %	43.2 %
Isaac B · 50 layouts	73.6 %	51.8 mm	29.0 %	48.6 %

- **The grasp rate is high.** Under Isaac M it is 87–90 %, higher than OFT (66–68 %) or the original OpenVLA (41–44 %) in the same cells.
- **Placement is less accurate.** In MuJoCo almost every success lands within 30 mm of the plate center (success 98.4 %, within 30 mm 98.4 %). Under Isaac M, success is 61 % but only 34–37 % lands within 30 mm, so many successes place the bowl toward the plate edge.
- **Under MuJoCo-B it still places on the plate, but less accurately.** Within 30 mm stays at 96.0 %, but the median placement error grows from 12.1 to 19.7 mm.

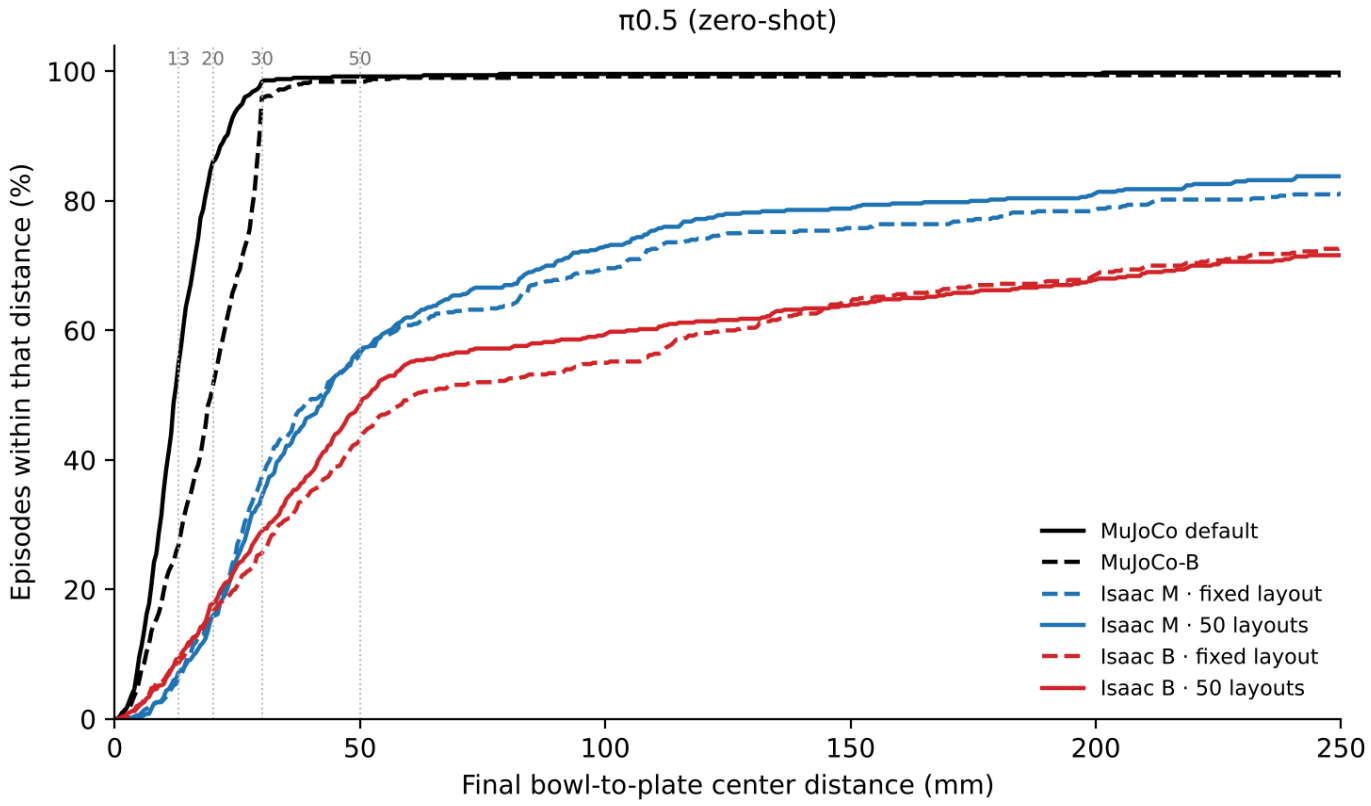


Figure 7. Cumulative distribution of the final bowl-to-plate center distance (including failed episodes).

9. What now appears to be the cause

- **Lifting.** Under M, gripping without lifting and regrasping are common. This points the same way as the measurement that lifting moves 0.63× of MuJoCo (being checked).
- **Where the gripper closes.** Under B, failures that overshoot the grasp position during the 5-action chunk increase.
- **Placement accuracy.** Even after bringing the bowl to the plate, it often places it near the edge or releases it from slightly high.

10. Limitations and next experiments

Limitations.

- Each cell was run once. Unverified successes have not yet been checked on video.

- In Isaac Sim the original criterion is the evaluation code's check; MuJoCo uses LIBERO's own check. One-pass grouping and failure kinds use the same rules in both simulators.
- The number of actions executed per chunk (5) follows the openpi example; other numbers were not tested.
- π0.5 was not retrained on Isaac Sim data.

Next experiments.

1. Video check of unverified successes, starting with the "not held at release" episodes under B and the "never lifted with the hand" episodes in tasks 8 and 9.
2. Measure again under M2, to see whether regrasping drops and one-pass successes rise.
3. Whether executing fewer actions per chunk (e.g. 2–3) reduces the drop under B.

11. How to reproduce

Only this section uses the names and values from the code.

Cell	Result folder
MuJoCo default	`MJ_Pi05Z_default_layout50`
MuJoCo-B	`MJ_Pi05Z_B_layout50`
Isaac M · fixed layout	`Pi05Z_M_layout1`
Isaac M · 50 layouts	`Pi05Z_M_layout50`
Isaac B · fixed layout	`Pi05Z_B_layout1`
Isaac B · 50 layouts	`Pi05Z_B_layout50`

- Policy server: run the following in the openpi Python environment. The checkpoint is openpi `pi05_libero`; change the number of actions executed per chunk with `--chunk`.

```
XLA_PYTHON_CLIENT_PREALLOCATE=false uv run python code/pi05/pi05_server.py --port 8000
```

- Isaac Sim evaluation: `code/eval/eval_zs_all_isaac_v6.sh` with `CFGBASE=libero_spatial_environment_oft` (the same connection format as OFT). M uses `EXTRA="--sim-steps-per-action 3 --convergence-gain 0.65"`; the fixed layout uses `FIXED_STATE=00`.
- MuJoCo evaluation: `code/eval/mujoco_eval_chunk.py --model-name pi05_libero`. For MuJoCo-B, set the action interval to 10 per second.
- Criteria, tables and figures: `code/report/make_report_assets_v2.py`. The rules are in `code/analysis/vla_signals.py`, `code/analysis/strict_outcomes.py` and `code/analysis/grasp_rule.json`.

Appendix A. Full tables

The tables in this appendix are generated by `code/report/make_report_assets_v2.py` from `data/results_2026-10-13/episodes_all.csv` and inserted unchanged. The same numbers are in `data/results_2026-10-13/tables/` as CSV.

π0.5 (zero-shot)

Success rate

Cell	Episodes	Success (original)	95% CI	One-pass	If all unverified are one-pass	Retained vs MuJoCo default (original / one-pass)
MuJoCo default	500	492 (98.4%)	96.9–99.2	461 (92.2%)	476 (95.2%)	—

Cell	Episodes	Success (original)	95% CI	One-pass	If all unverified are one-pass	Retained vs MuJoCo default (original / one-pass)
MuJoCo-B	500	479 (95.8%)	93.7–97.2	435 (87.0%)	451 (90.2%)	97 % / 94 %
Isaac M · fixed layout	500	305 (61.0%)	56.7–65.2	150 (30.0%)	163 (32.6%)	62 % / 33 %
Isaac M · 50 layouts	500	303 (60.6%)	56.3–64.8	173 (34.6%)	181 (36.2%)	62 % / 38 %
Isaac B · fixed layout	500	260 (52.0%)	47.6–56.3	182 (36.4%)	198 (39.6%)	53 % / 39 %
Isaac B · 50 layouts	500	283 (56.6%)	52.2–60.9	183 (36.6%)	195 (39.0%)	58 % / 40 %

One-pass success (bowl carried without being lost)

Cell	Success (original)	One-pass success	Success after regrasp	Unverified success (video check)	Failure after a lost grasp	Other failure	Mean closings	Mean touching attempts
MuJoCo default	492	461 (92.2%)	16	15	2	6	1.06	1.05
MuJoCo-B	479	435 (87.0%)	28	16	10	11	1.24	1.16
Isaac M · fixed layout	305	150 (30.0%)	142	13	178	17	2.87	2.46
Isaac M · 50 layouts	303	173 (34.6%)	122	8	175	22	2.80	2.41
Isaac B · fixed layout	260	182 (36.4%)	62	16	207	33	2.88	2.36
Isaac B · 50 layouts	283	183 (36.6%)	88	12	179	38	2.77	2.32

Unverified successes: reasons and release height

Cell	Unverified	1 never lifted with the hand	2 no on-plate moment	3 not held at release	3: <5 cm	5–10 cm	≥10 cm	never held	Tasks with reason 1 (task:count)
MuJoCo default	15	0	1	14	5	4	5	0	—
MuJoCo-B	16	0	5	11	1	5	5	0	—
Isaac M · fixed layout	13	5	0	8	3	5	0	0	8:3 9:2
Isaac M · 50 layouts	8	4	0	4	1	2	1	0	8:3 9:1
Isaac B · fixed layout	16	1	1	14	1	4	9	0	3:1
Isaac B · 50 layouts	12	1	1	10	1	2	7	0	8:1

Failure kinds

Cell	Success	Grasped but not lifted	Closed without grasping	Never closed	Lifted, not carried to plate	Failed near plate
MuJoCo default	492	1	0	0	1	6
MuJoCo-B	479	0	3	0	1	17

Cell	Success	Grasped but not lifted	Closed without grasping	Never closed	Lifted, not carried to plate	Failed near plate
Isaac M · fixed layout	305	72	64	0	26	33
Isaac M · 50 layouts	303	60	45	0	39	53
Isaac B · fixed layout	260	29	125	0	71	15
Isaac B · 50 layouts	283	26	132	0	46	13

Detailed metrics (Table 1 metrics)

Cell	Gripper close %	Grasp %	Placement %	Undisturbed %	Median error (mm)	≤13 mm %	≤20 mm %	≤30 mm %	≤50 mm %
MuJoCo default	100.0	100.0	98.4	91.6	12.1	54.8	86.0	98.4	99.2
MuJoCo-B	100.0	99.2	95.8	92.8	19.7	26.8	51.4	96.0	98.4
Isaac M · fixed layout	100.0	86.8	61.0	—	41.9	6.4	15.6	37.4	56.4
Isaac M · 50 layouts	100.0	90.0	60.6	—	42.9	7.2	16.0	34.4	57.0
Isaac B · fixed layout	100.0	74.2	52.0	—	61.0	8.6	16.6	25.6	43.2
Isaac B · 50 layouts	100.0	73.6	56.6	—	51.8	9.2	17.8	29.0	48.6

Undisturbed counts only episodes that log other objects' positions (currently MuJoCo only). Distance metrics include failed episodes.

Per-task success (original criterion)

Task	Bowl location	MuJoCo default	MuJoCo-B	Isaac M · fixed layout	Isaac M · 50 layouts	Isaac B · fixed layout	Isaac B · 50 layouts
0	between plate and ramekin	50	48	42	43	14	22
1	next to ramekin	50	45	32	23	20	34
2	table center	50	50	25	21	42	49
3	on cookie box	48	49	48	45	31	34
4	in cabinet top drawer	49	47	7	18	30	34
5	on ramekin	49	46	34	35	1	0
6	next to cookie box	49	50	46	41	50	41
7	on stove	50	50	0	0	7	8
8	next to plate	49	47	32	36	44	45
9	on wooden cabinet	48	47	39	41	21	16
Total		492	479	305	303	260	283

Per-task one-pass successes

Task	Bowl location	MuJoCo default	MuJoCo-B	Isaac M · fixed layout	Isaac M · 50 layouts	Isaac B · fixed layout	Isaac B · 50 layouts
0	between plate and ramekin	41	41	14	22	0	5
1	next to ramekin	49	36	8	8	11	27

Task	Bowl location	MuJoCo default	MuJoCo-B	Isaac M · fixed layout	Isaac M · 50 layouts	Isaac B · fixed layout	Isaac B · 50 layouts
2	table center	49	46	4	3	38	44
3	on cookie box	48	48	41	43	10	11
4	in cabinet top drawer	49	46	7	15	30	32
5	on ramekin	48	42	22	12	0	0
6	next to cookie box	48	48	42	33	47	21
7	on stove	41	49	0	0	4	3
8	next to plate	44	39	3	13	30	29
9	on wooden cabinet	44	40	9	24	12	11
Total		461	435	150	173	182	183

One factor changed at a time

Comparison	Success (original) Difference now (z)	One-pass Difference now (z)	One-pass Possible range after video check (z)
Isaac M · fixed layout → Isaac B · fixed layout	-9.0 (-2.87)	+6.4 (+2.15)	+3.8 ~ +9.6 (+1.26 ~ +3.19)
Isaac M · 50 layouts → Isaac B · 50 layouts	-4.0 (-1.28)	+2.0 (+0.66)	+0.4 ~ +4.4 (+0.13 ~ +1.44)
MuJoCo default → MuJoCo-B	-2.6 (-2.45)	-5.2 (-2.69)	-8.2 ~ -2.0 (-4.55 ~ -1.12)
Isaac M · fixed layout → Isaac M · 50 layouts	-0.4 (-0.13)	+4.6 (+1.56)	+2.0 ~ +6.2 (+0.67 ~ +2.08)
Isaac B · fixed layout → Isaac B · 50 layouts	+4.6 (+1.46)	+0.2 (+0.07)	-3.0 ~ +2.6 (-0.98 ~ +0.85)
MuJoCo default → Isaac M · 50 layouts	-37.8 (-14.80)	-57.6 (-18.91)	-60.6 ~ -56.0 (-20.08 ~ -18.47)
MuJoCo default → Isaac B · 50 layouts	-41.8 (-15.83)	-55.6 (-18.36)	-58.6 ~ -53.2 (-19.55 ~ -17.71)

Appendix B. Terms

Term	Meaning
Episode	One trial. 500 per cell (10 tasks × 50)
Action · chunk	An action is a short movement command the model issues. π0.5 issues 10 actions per observation; in these tests the first 5 are executed before it looks again
Zero-shot	Testing in the target environment (Isaac Sim) without additional training on its data
M / B	Isaac Sim execution settings. M executes each action for 0.05 s with the hand target at 65 % (MuJoCo-matched speed). B executes for 0.17 s at the full commanded distance
MuJoCo-B	MuJoCo with the action interval lengthened to 10 per second (0.10 s)
Fixed layout / 50 layouts	Objects always at LIBERO's first placement / one of 50 placements per episode
One-pass success	The bowl was picked up and placed on the plate without ever being lost or dropped
Success after regrasp	The bowl was touched and lost (including gripped but not lifted) or dropped, then grasped again and placed
Unverified success	Success by the original criterion, but the logs alone cannot confirm that the bowl was picked up and carried; needs a video check
z	Tells whether a difference between two success rates could be chance. Above 1.96 in absolute value means less than 5 % chance